BLACK FRIDAY SALES ANALYTICS REPORT

Customer Behavior, Segmentation & Product Intelligence

Dataset Black Friday Retail Transactions	**Records** 550,068 Transactions | 5,891 Customers
Tools Used Python (Pandas, Seaborn, Matplotlib) | Power BI	**Total Revenue** ₹5.10 Billion

Prepared for Academic & Business Intelligence Purposes

Executive Summary

This report presents a comprehensive end-to-end data analytics study of Black Friday retail transaction data, encompassing over 550,000 transactions across 5,891 unique customers and 3,631 products. The analysis integrates Python-based Exploratory Data Analysis (EDA) with interactive Power BI dashboards to derive actionable business intelligence.

Project Scope & Objectives

- Analyze customer purchasing behavior using Black Friday transaction data
- Identify and segment customers into high-value and low-value groups
- Evaluate the impact of demographic variables (age, gender, city category, marital status)
- Assess product category performance and revenue contribution
- Develop interactive Power BI dashboards for visual decision support
- Generate data-driven recommendations for marketing and revenue optimization

Key Performance Highlights

₹5.10 Bn	**5,891**	**550,068**	**₹9,264**
Total Revenue	Total Customers	Total Transactions	Avg Purchase/Txn

1,179	**55.4%**	**3,631**	**26–35**
High-Value Customers	HV Revenue Share	Unique Products	Top Age Segment

Critical Findings at a Glance

Pareto Revenue Concentration

Approximately 20% of customers (1,179 high-value customers) contribute over 55% of total revenue (₹2.82 Bn out of ₹5.10 Bn), confirming a strong Pareto distribution in this retail dataset.

Dominant Customer Demographic

The 26–35 age group is the single most valuable segment, generating ₹2.03 Bn (39.8% of total revenue) — nearly double the next segment (36–45 at ₹1.03 Bn).

Gender Spending Gap

Male customers account for 75.3% of all transactions and generate ₹3.91 Bn in revenue. The average male spend per transaction (₹9,438) is statistically significantly higher than female spend (₹8,735).

Product Category Concentration

Product Category 1 alone drives ₹1.91 Bn (37.5% of total revenue), followed by Category 5 (₹941.8 M) and Category 8 (₹854.3 M). The top 3 categories contribute over 70% of revenue.

Table of Contents

Chapter 1: Introduction

1.1 Background

In the modern retail landscape, data has emerged as one of the most valuable assets for organizations seeking competitive advantage. Black Friday — one of the world's largest annual retail events — generates an extraordinary volume of transactions within a compressed timeframe, creating a uniquely rich environment for behavioral and commercial analysis.

With millions of transactions occurring in a single event window, Black Friday data offers unparalleled opportunities to study consumer decision-making, demographic influences on spending, and product category demand patterns. Leveraging advanced analytics tools such as Python and Power BI, organizations can now transform this raw transactional data into strategic intelligence that drives tangible business outcomes.

1.2 Problem Statement

Despite access to extensive transactional datasets, many retail organizations continue to struggle with converting raw data into actionable insights. Key business challenges include:

- Identifying which customers contribute the highest value and how to retain them
- Understanding how demographic factors (age, gender, geographic location) shape purchasing behavior
- Determining which product categories and SKUs drive the highest revenue and premium engagement
- Analyzing repeat purchase behavior to inform customer loyalty and retention programs
- Communicating complex analytical findings to non-technical business stakeholders

Without structured analytical frameworks, organizations risk making strategic decisions based on intuition rather than data, leading to missed revenue opportunities and inefficient resource allocation.

1.3 Objectives of the Study

This project is designed to achieve the following primary objectives:

- To conduct a comprehensive exploratory data analysis (EDA) of Black Friday transaction data
- To segment customers into high-value and low-value groups based on total spending
- To quantify and explain the influence of demographic factors on purchase behavior
- To evaluate product and category-level performance and revenue contribution
- To study repeat customer behavior and identify patterns that support retention strategies
- To develop a multi-page interactive Power BI dashboard for executive-level reporting
- To formulate specific, data-backed business recommendations

1.4 Scope of the Study

This study analyzes a historical Black Friday retail transaction dataset containing 550,068 records and 12 variables. The scope includes customer-level aggregation, statistical hypothesis testing, demographic profiling, and product performance evaluation. The analysis is confined to the variables available within the dataset — external data sources such as pricing information, promotional campaign data, or time-series trends are not incorporated.

Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy) is used for all data processing and statistical analysis. Power BI is used for interactive dashboard development and business presentation.

1.5 Report Structure

This report is organized across the following chapters:

Chapter	Title	Focus Area
1	Introduction	Context, objectives, scope
2	Literature Review	Theoretical foundations
3	Dataset Description	Data variables and characteristics
4	Data Preprocessing	Cleaning and feature engineering
5	Methodology	Analytical approach and tools
6	Exploratory Data Analysis	Statistical and visual insights
7	Customer Segmentation	High-value vs. low-value analysis
8	Product & Category Analysis	Revenue by product/category
9	Power BI Dashboard	Visualization and reporting
10	Results & Findings	Summary of key insights
11	Discussion	Interpretation and business implications
12	Recommendations	Data-backed strategic actions
13	Conclusion	Final summary

Chapter 2: Literature Review

2.1 Retail Analytics

Retail analytics is defined as the systematic use of data and statistical methods to understand and improve retail performance. In an increasingly data-rich environment, retailers leverage analytics to gain insight into customer behavior, streamline inventory management, and develop precision marketing strategies.

The rise of e-commerce and large-scale transaction processing has democratized access to behavioral data at scale. Techniques such as descriptive statistics, cluster analysis, and predictive modeling have become standard components of modern retail intelligence frameworks.

2.2 Customer Segmentation Theory

Customer segmentation is a cornerstone of modern marketing science, enabling organizations to identify and engage distinct customer groups with tailored strategies. Value-based segmentation — classifying customers by their total economic contribution — is among the most practically impactful approaches.

RFM (Recency, Frequency, Monetary) analysis is a widely adopted segmentation framework in retail, measuring how recently a customer purchased, how often, and how much they spend. For this study, a monetary-value threshold approach is used to define high-value customers, consistent with academic and industry practice.

2.3 The Pareto Principle in Retail

The Pareto Principle — commonly stated as the '80/20 rule' — posits that a disproportionate share of effects is generated by a minority of causes. In retail analytics, this typically manifests as a small percentage of customers driving a large majority of revenue.

Empirical research across retail sectors consistently validates this principle. Organizations that identify and strategically invest in their high-value customer cohort regularly outperform those that apply uniform treatment across their entire customer base. This project's findings strongly confirm the Pareto distribution in the Black Friday dataset.

2.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis, introduced by statistician John Tukey, is an investigative approach to data examination that prioritizes pattern discovery, anomaly detection, and hypothesis generation. EDA uses visual techniques (histograms, box plots, scatter plots) and summary statistics to surface meaningful structure in data before formal modeling.

In this project, EDA forms the analytical backbone, enabling systematic investigation of purchase distributions, demographic comparisons, and category-level patterns. Python libraries including Pandas, Seaborn, and Matplotlib are employed throughout.

2.5 Statistical Hypothesis Testing

Statistical hypothesis testing provides a rigorous framework for determining whether observed differences between groups are likely to reflect genuine population-level effects or could plausibly arise from chance. Given the non-normal distribution of purchase amounts (right-skewed), non-parametric tests are preferred:

- Mann-Whitney U Test: for comparing two independent groups (e.g., Male vs. Female)
- Kruskal-Wallis Test: for comparing three or more independent groups (e.g., Age groups, City categories)

A significance level of $\alpha = 0.05$ is applied throughout. Results with $p\text{-value} < 0.05$ are considered statistically significant.

2.6 Business Intelligence and Data Visualization

Business Intelligence (BI) platforms such as Microsoft Power BI enable the transformation of analytical outputs into interactive, stakeholder-facing dashboards. Effective BI design emphasizes clarity, contextual KPIs, and actionable insight delivery over data volume.

This project's Power BI dashboard is designed across four pages — Sales Performance Overview, Customer Segmentation, Product & Category Intelligence, and Strategy & Actionable Insights — reflecting best practices in executive reporting.

Chapter 3: Dataset Description

3.1 Dataset Overview

The dataset used in this project represents transactional records from a Black Friday retail event, provided for academic analytical purposes. It contains 550,068 individual purchase transactions across 5,891 unique customers and 3,631 distinct products.

550,068	5,891	3,631	12
Total Records	Unique Customers	Unique Products	Original Columns

3.2 Variable Descriptions

Variable	Type	Description	Notes
User_ID	Numerical	Unique customer identifier	No missing values
Product_ID	Categorical	Unique product identifier	No missing values
Gender	Categorical	Customer gender (M/F)	75.3% Male, 24.7% Female
Age	Categorical	Age group (7 bands)	26–35 is dominant group
Occupation	Numerical	Occupation code (0–20)	Coded; not descriptive
City_Category	Categorical	City tier: A, B, or C	B = 42%, C = 31%, A = 27%
Stay_In_Current_City_Years	Categorical	Tenure in current city	Ranges: 0 to 4+
Marital_Status	Binary	0 = Single, 1 = Married	59% Single, 41% Married
Product_Category_1	Numerical	Primary product category (1–20)	Always present
Product_Category_2	Numerical	Secondary product category	31.6% missing
Product_Category_3	Numerical	Tertiary product category	69.7% missing
Purchase	Numerical	Transaction amount in INR	Target variable

3.3 Target Variable: Purchase

The Purchase variable represents the monetary value of each transaction in Indian Rupees (INR) and serves as the primary analytical focus across all chapters. Its statistical distribution is as follows:

Statistic	Value
Minimum	₹12

25th Percentile (Q1)	₹5,823
Median (Q2)	₹8,047
Mean	₹9,264
75th Percentile (Q3)	₹12,054
Maximum	₹23,961
Standard Deviation	₹5,023
Total Revenue	₹5,095,812,742

The distribution is right-skewed, with the mean (₹9,264) substantially higher than the median (₹8,047), indicating the presence of high-value transactions pulling the average upward. This skewness motivates the use of non-parametric statistical tests throughout the analysis.

3.4 Missing Value Profile

Two variables contain significant missing values that require deliberate treatment:

Variable	Missing Count	Missing %	Treatment Applied
Product_Category_1	0	0.0%	No treatment required
Product_Category_2	173,638	31.6%	Indicator variable (PC2_missing) created
Product_Category_3	383,247	69.7%	Indicator variable (PC3_missing) created
All other variables	0	0.0%	Complete data; no imputation needed

Chapter 4: Data Cleaning & Preprocessing

4.1 Overview of Preprocessing Pipeline

Data preprocessing is the critical bridge between raw transactional data and reliable analytical outputs. In this project, the preprocessing pipeline was designed to preserve maximum information while ensuring data quality and analytical integrity. The pipeline comprises four stages: missing value treatment, outlier analysis, feature engineering, and data type validation.

4.2 Missing Value Treatment

Product_Category_2 and Product_Category_3 contain substantial proportions of missing values (31.6% and 69.7% respectively). Rather than removing these records — which would have resulted in a loss of over 380,000 rows — binary indicator variables were created to preserve missingness as meaningful information.

- PC2_missing: Binary flag (1 = Product_Category_2 was missing, 0 = present)
- PC3_missing: Binary flag (1 = Product_Category_3 was missing, 0 = present)

This approach preserves all 550,068 records while enabling downstream analysis of how category availability relates to purchase behavior.

4.3 Outlier Analysis

The Purchase variable was examined for outliers using the Interquartile Range (IQR) method. The IQR (Q3 - Q1 = 12,054 - 5,823 = 6,231) yields theoretical boundaries of [5,823 - 1.5×6,231, 12,054 + 1.5×6,231] = [-3,523, 21,401]. Values above ₹21,401 represent statistical outliers.

However, upon inspection, these high-value transactions (up to ₹23,961) are genuine premium purchases rather than data errors. They are retained in the analysis to avoid understating the contribution of high-spending customers, but their influence is noted in relevant statistical comparisons.

4.4 Feature Engineering

Two derived features were created to enhance analytical utility:

Derived Feature	Definition	Purpose
Metro_Group	City_Category A = Metro; B & C = Non-Metro	Simplifies geographic analysis into a binary dimension
Customer_Total_Purchase	Sum of all purchases per User_ID	Enables customer-level segmentation and value scoring
PC2_missing	1 if Product_Category_2 is null, else 0	Preserves missingness as an analytical variable
PC3_missing	1 if Product_Category_3 is null, else 0	Preserves missingness as an analytical variable

4.5 Data Type Validation

All variables were validated for appropriate data types prior to analysis. Categorical variables (Gender, Age, City_Category, Stay_In_Current_City_Years) were verified as string/object types; numerical variables (Purchase, Occupation, Product_Category_1, Marital_Status) were confirmed as integer or float types. No type mismatches or encoding errors were identified.

4.6 Preprocessing Summary

Final Dataset State

After preprocessing: 550,068 records retained (0% row loss), 16 analytical features (12 original + 4 derived), zero null values in the target variable (Purchase), and two binary indicator variables preserving missing category information. The dataset is complete, consistent, and analytically ready.

Chapter 5: Methodology

5.1 Analytical Framework

This project adopts a structured, multi-stage analytical framework combining descriptive statistics, inferential testing, customer segmentation, and business intelligence visualization. The framework is designed to move systematically from data understanding to business insight generation.

5.2 Analytical Stages

Stage	Technique	Tool	Objective
1 – Data Profiling	Descriptive statistics	Python / Pandas	Understand distributions and data quality
2 – EDA	Histograms, box plots, bar charts	Matplotlib / Seaborn	Identify patterns across demographics
3 – Statistical Testing	Mann-Whitney U, Kruskal-Wallis	SciPy	Validate group differences statistically
4 – Segmentation	Percentile-based value scoring	Python / Pandas	Classify customers as High/Low value
5 – Product Analysis	Revenue aggregation by category/SKU	Python / Pandas	Identify top-performing products
6 – Visualization	Interactive multi-page dashboard	Power BI / DAX	Communicate insights to stakeholders

5.3 Customer Segmentation Methodology

Customer segmentation is performed at the user level by aggregating total spend (Customer_Total_Purchase). The 80th percentile of total customer spend serves as the segmentation threshold:

- High-Value Customers: Total spend >= 80th percentile threshold
- Low-Value Customers: Total spend < 80th percentile threshold

This approach classifies approximately 20% of customers as high-value, consistent with the Pareto Principle and widely used in RFM-based segmentation practice. The threshold was validated by comparing revenue contribution across segments.

5.4 Statistical Testing Protocol

All statistical tests are conducted at the 5% significance level ($\alpha = 0.05$). Non-parametric tests are selected given the right-skewed, non-normally distributed purchase data:

Comparison	Test Applied	Null Hypothesis
Male vs. Female spend	Mann-Whitney U	No difference in purchase distributions
Age group spend	Kruskal-Wallis	No difference across age groups
City category spend	Kruskal-Wallis	No difference across City A, B, C

Marital status spend	Mann-Whitney U	No difference between Single and Married

5.5 Tools & Technologies

- Python 3.x with libraries: Pandas (data manipulation), NumPy (numerical operations), Matplotlib & Seaborn (visualization), SciPy (statistical testing)
- Microsoft Power BI Desktop: Interactive dashboard development using DAX measures
- Microsoft Word / Docx: Report documentation and delivery

Chapter 6: Exploratory Data Analysis

6.1 Purchase Amount Distribution

The distribution of individual transaction amounts spans from ₹12 to ₹23,961, with a median of ₹8,047 and mean of ₹9,264. The histogram reveals a multi-modal distribution with distinct peaks, suggesting that products cluster into price bands. This multi-modality indicates the dataset captures transactions across multiple product tiers simultaneously.

Distribution Insight

The right-skewed, multi-modal distribution of purchase amounts confirms that retail spending is not uniformly distributed. Multiple price-point clusters — evident in the histogram — suggest distinct product category price ranges rather than a single continuous price spectrum.

6.2 Gender Analysis

Male customers constitute 75.3% of all transactions (414,259) and generate ₹3.91 Bn (76.7%) of total revenue. Female customers account for 24.7% of transactions and ₹1.19 Bn (23.3%) of revenue.

Gender	Transactions	Transaction %	Revenue (₹)	Revenue %	Avg Purchase (₹)
Male	414,259	75.3%	3,909,580,100	76.7%	9,438
Female	135,809	24.7%	1,186,232,642	23.3%	8,735

The Mann-Whitney U test confirms a statistically significant difference in purchase distributions between male and female customers ($p < 0.0001$). While the absolute spend gap is modest (₹703 per transaction), it compounds significantly across 400,000+ male transactions.

6.3 Age Group Analysis

Age group is among the most discriminating variables in this dataset. The 26–35 cohort dominates in both transaction volume and revenue generation, followed by the 36–45 age group.

Age Group	Transactions	Revenue (₹)	Revenue %	Avg Purchase (₹)
0–17	15,102	134,913,183	2.6%	8,933
18–25	99,660	913,848,675	17.9%	9,170
26–35	219,587	2,031,770,578	39.9%	9,252
36–45	110,013	1,026,569,884	20.1%	9,331
46–50	45,701	420,843,403	8.3%	9,209
51–55	38,501	367,099,644	7.2%	9,535
55+	21,504	200,767,375	3.9%	9,336

The Kruskal-Wallis test confirms statistically significant differences in purchase behavior across age groups ($p < 0.0001$). The 26–35 group generates ₹2.03 Bn — more than the 36–45 and 18–25 groups combined.

6.4 City Category Analysis

City Category	Transactions	Revenue (₹)	Revenue %	Avg Purchase (₹)
A (Metro)	147,720	1,316,471,661	25.8%	8,912
B	231,173	2,115,533,605	41.5%	9,151
C	171,175	1,663,807,476	32.6%	9,720

City C customers exhibit the highest average purchase (₹9,720), while City B contributes the greatest total revenue by volume. The Kruskal-Wallis test indicates statistically significant differences across city categories ($p < 0.0001$).

6.5 Marital Status Analysis

Single customers (Marital_Status = 0) account for 59% of transactions, while married customers account for 41%. The Mann-Whitney U test reveals no statistically significant difference in purchase amounts between single and married customers ($p = 0.408$), indicating that marital status does not meaningfully influence spending behavior in this dataset.

Key Finding

Marital status is the only demographic variable tested that does NOT show a statistically significant relationship with purchase amounts ($p = 0.408$). Businesses should not prioritize marital-status-based segmentation for this retail context.

Chapter 7: Customer Segmentation Analysis

7.1 Segmentation Overview

Customer segmentation is performed at the user level using total spend as the primary segmentation criterion. The 80th percentile of Customer_Total_Purchase serves as the threshold dividing High-Value (HV) and Low-Value (LV) segments.

1,179	4,712	2.82 Bn	55.4%
High-Value Customers	Low-Value Customers	HV Revenue (₹)	HV Revenue Share

Pareto Confirmation

The 20% of customers classified as High-Value contribute 55.4% of total revenue — confirming a stronger-than-classic Pareto relationship. This revenue concentration effect underscores the critical importance of HV customer retention and engagement strategies.

7.2 High-Value Customer Demographics

Examining the demographic profile of High-Value customers reveals concentrated characteristics that can guide targeted marketing:

Dimension	High-Value Characteristics	Strategic Implication
Age	Predominantly 26–35 (513), followed by 36–45 (245)	Target working professionals aged 26–45
Gender	Male customers overrepresented in HV segment	Gender-aware premium product marketing
City Category	City C shows highest HV revenue rate	Non-metro premium segments are underserved
Revenue per Customer	₹2.82 Bn across 1,179 customers	Average HV customer value ≈ ₹2.39 M per event

7.3 High-Value by Age Group

Age Group	HV Count	HV Revenue (₹)	Revenue Contribution
26–35	513	1,000,000,000+	Largest HV cohort by count and revenue
36–45	245	~650,000,000	Second largest HV cohort
18–25	219	~450,000,000	Third largest — opportunity for growth
46–50	79	~280,000,000	Smaller but high-value per customer

51–55	77	~260,000,000	Loyal older segment
55+	26	~110,000,000	Smallest — limited but stable
0–17	20	~70,000,000	Minimal HV presence

7.4 Revenue Split: HV vs. LV Customers

Segment	Customer Count	Customer %	Revenue (₹)	Revenue %	Avg Revenue/Customer
High-Value	1,179	20.0%	2,821,000,351	55.4%	₹2,392,706
Low-Value	4,712	80.0%	2,274,812,391	44.6%	₹482,747

The average High-Value customer generates approximately 5 times the revenue of a Low-Value customer (₹2.39 M vs ₹0.48 M). This magnitude of difference justifies significant investment in HV customer acquisition, retention, and personalized engagement programs.

7.5 Repeat Customer Behavior

All 5,891 customers in the dataset have made multiple transactions (minimum 6 per customer), with the median customer making 54 transactions and a maximum of 1,026. This indicates a dataset captured over an extended window rather than a single-day event, enriching behavioral analysis.

Percentile	Transactions per Customer
Minimum	6
25th Percentile	26
Median (50th)	54
75th Percentile	117
Maximum	1,026

Chapter 8: Product & Category Analysis

8.1 Product Category Performance Overview

The dataset includes three product category dimensions (Product_Category_1, Product_Category_2, Product_Category_3), with Category 1 being universally present. Revenue analysis by Product_Category_1 reveals a highly concentrated demand structure.

8.2 Top Product Categories by Revenue

Category	Total Revenue (₹)	Revenue %	Transaction Volume
Category 1	1,910,013,754	37.5%	Highest volume category
Category 5	941,835,229	18.5%	Second largest
Category 8	854,318,799	16.8%	Third largest
Category 6	324,150,302	6.4%	Mid-tier performer
Category 2	268,516,186	5.3%	Moderate volume
Category 3	204,084,713	4.0%	Moderate volume
Category 16	145,120,612	2.8%	Niche segment
All Others	447,773,147	8.8%	Combined remaining categories

Concentration Finding

The top 3 product categories (Categories 1, 5, and 8) collectively generate ₹3.71 Bn, representing 72.8% of total revenue. This extreme concentration indicates that business strategy should prioritize depth in these categories over breadth across the full range of 20 categories.

8.3 Top 10 Products by Total Revenue

Rank	Product ID	Total Revenue (₹)	Revenue Tier
1	P00025442	27,995,166	Top performer
2	P00110742	26,722,309	Top performer
3	P00255842	25,168,963	Top performer
4	P00059442	24,338,343	High performer
5	P00184942	24,334,887	High performer
6	P00112142	24,244,000+	High performer
7	P00110942	23,618,000+	High performer
8	P00237542	23,414,000+	High performer
9	P00057642	23,102,000+	High performer

10	P00010742	22,248,000+	High performer

The top product P00025442 alone generates ₹28.0 M in revenue. Notably, even the 10th-ranked product generates over ₹22 M, indicating a competitive top-tier SKU environment. Investment in premium product promotions for these items would yield disproportionate returns.

8.4 High-Value Revenue Penetration by Category

High-Value revenue penetration (the proportion of each category's revenue generated by HV customers) identifies which categories are premium-engagement drivers:

Category	HV Revenue Penetration %	Classification
Category 18	62%	Premium Category — Highest HV penetration
Category 9	61%	Premium Category
Category 11	59%	Premium Category
Category 17	58%	Near-Premium
Category 7	58%	Near-Premium
Category 1	~55%	Volume + Value balanced
Category 5	~51%	Volume-driven category

Categories 18, 9, and 11 show the highest premium penetration (>59%), making them ideal candidates for targeted premium marketing, exclusive membership offers, and high-value loyalty incentives.

8.5 Top Occupations by Revenue

Occupation Code	Total Revenue (₹)	Cluster Assignment
4	666,244,484	Cluster 2 (Highest)
0	635,406,958	Cluster 2
7	557,371,587	Cluster 4
1	424,614,144	Cluster 3
17	393,281,453	Cluster 3

Chapter 9: Power BI Dashboard Development

9.1 Dashboard Architecture

The Power BI dashboard is designed as a multi-page analytical interface providing progressive depth — from executive overview to granular operational insights. Each page is architecturally distinct, optimized for a specific stakeholder audience and decision context.

Dashboard Page	Title	Primary Audience	Key Purpose
Page 1	Sales Performance Overview	C-suite / Executive	Top-line KPIs and sales distribution
Page 2	Customer Segmentation & Behavior	Marketing / CRM	HV/LV analysis, repeat patterns
Page 3	Product & Category Intelligence	Product / Merchandising	SKU and category revenue analysis
Page 4	Strategy & Actionable Insights	Strategy / Operations	Data-backed recommendations

9.2 Page 1: Sales Performance Overview

The overview page presents six headline KPIs in a card-based layout: Total Sales (₹5.10 Bn), Total Customers (6K), Average Purchase per Customer (₹865.02K), Premium Segment Size (1K), and High-Value Contribution % (0.55).

Supporting visuals include: a bar chart showing Sales by Age group, a donut chart for City_Category revenue share (B: 41.52%, C: 32.65%, A: 25.83%), a marital status revenue split donut, and a ranked table of top occupations by revenue and cluster assignment.

9.3 Page 2: Customer Segmentation & Behavior

This page operationalizes the customer segmentation analysis with the following visual components:

- Bar chart: High-Value Customers by Age group — 26–35 dominates with 513 HV customers
- Bar chart: Repeat Customers by Age group — 26–35 leads with 2.1K repeat customers
- Donut: High-Value Revenue by City Category — City A contributes 34.0%, City C 54.4%
- Bar chart: Customer Count by Segment (LowValue: 4.7K, HighValue: 1.2K)
- Stacked bar: HV vs. LV Revenue split (55.36% vs. 44.64%)
- KPI cards: HV Revenue (₹2.82 Bn), LV Revenue (₹2.27 Bn), HV Rate (0.20), Avg Purchase/Customer (₹865.02K)

9.4 Page 3: Product & Category Intelligence

The product page centers on revenue contribution by SKU and category, featuring:

- Ranked bar chart: Top 10 Products by Total Revenue (P00025442 leading at ₹28.0M)
- Grouped bar chart: High-Value Revenue Penetration % by Product Category (Categories 18, 9, 11 at top)
- Ranked bar chart: Top 10 Products by Average Revenue — Category 10 products lead at ₹4.0M average

- Product Category grid (1–20): Color-coded to distinguish high and low performers

9.5 Page 4: Strategy & Actionable Insights

The strategy page synthesizes the analytical findings into three structured insight columns — Customer Insight, Product Insight, and Behavior Insight — accompanied by three executive-level recommendations with specific, measurable action items.

9.6 DAX Measures & Calculated Columns

The following custom DAX measures power the dashboard calculations:

- Total_Revenue = SUM(data[Purchase]) — baseline revenue aggregation
- HighValue_Revenue = CALCULATE([Total_Revenue], data[Segment]='HighValue') — segment-filtered revenue
- HighValue_Rate = DIVIDE([HighValue_Cust], [Total_Cust]) — premium customer ratio
- Avg_Purchase_Per_Customer = DIVIDE([Total_Revenue], [Total_Customers]) — customer-level average
- HV_Revenue_Pct = DIVIDE([HighValue_Revenue], [Total_Revenue]) — contribution share

Chapter 10: Results & Findings

10.1 Customer Segmentation Results

The segmentation analysis produces a clear bifurcated customer structure confirming Pareto-like revenue concentration:

Finding	Metric	Business Implication
HV customers = 20% of base	1,179 of 5,891	Focus retention effort on 1-in-5 customers
HV revenue = 55.4% of total	₹2.82 Bn of ₹5.10 Bn	Over half of revenue is at risk without HV retention
HV avg revenue 5x LV avg	₹2.39M vs ₹0.48M per customer	Acquisition cost justified at 5x premium
26–35 = 43.5% of HV customers	513 of 1,179 HV customers	Primary target cohort for premium programs

10.2 Demographic Findings Summary

Variable	Test	p-value	Significant?	Key Finding
Gender	Mann-Whitney U	< 0.0001	Yes	Males spend ₹703 more per transaction on average
Age	Kruskal-Wallis	< 0.0001	Yes	26–35 is dominant group; 39.9% of revenue
City Category	Kruskal-Wallis	< 0.0001	Yes	City C has highest avg purchase (₹9,720)
Marital Status	Mann-Whitney U	0.408	No	No spending difference between single/married

10.3 Product & Category Findings

- Category 1 dominates with ₹1.91 Bn (37.5% of all revenue) — a critical business pillar
- Top 3 categories (1, 5, 8) account for 72.8% of revenue — extreme concentration risk and opportunity
- Categories 18, 9, and 11 exhibit the highest HV penetration (>59%) — premium segment anchors
- Top product P00025442 generates ₹28.0 M — a flagship SKU warranting priority investment
- Top 10 products generate approximately ₹250 M combined — 4.9% of revenue from 0.3% of SKUs

10.4 Statistical Summary Table

Analysis Area	Method	Result	Confidence Level

Gender spend difference	Mann-Whitney U	Statistically significant	99.9%+
Age group spend variation	Kruskal-Wallis	Statistically significant	99.9%+
City category spend variation	Kruskal-Wallis	Statistically significant	99.9%+
Marital status spend	Mann-Whitney U	Not significant	59.2% (below threshold)
Revenue concentration (HV)	Percentile analysis	Top 20% = 55.4% revenue	Descriptive
Product concentration	Revenue aggregation	Top 3 cat. = 72.8% revenue	Descriptive

Chapter 11: Discussion

11.1 Revenue Concentration and the Pareto Effect

The most compelling structural finding of this analysis is the revenue concentration among High-Value customers. With 20% of customers generating 55.4% of revenue, the dataset exhibits a Pareto relationship — and in fact slightly exceeds the classical 80/20 expectation (20% generating 55%, not 80%).

This implies that the retailer's revenue sustainability is heavily dependent on the engagement and retention of a small customer cohort. A single-digit percentage decline in HV customer activity could translate to a double-digit revenue impact at the organizational level. This finding argues strongly for a tiered service and retention model that dedicates disproportionate resources to the HV segment.

11.2 Demographic Influences: Age as the Dominant Variable

Of all demographic variables tested, age emerges as the most powerful predictor of revenue contribution at the aggregate level. The 26–35 age group, likely representing early-to-mid career professionals with growing disposable income, generates ₹2.03 Bn — nearly 40% of total revenue despite representing only 39.9% of transactions.

The practical implication is that marketing budgets, product curation, and loyalty investment should be concentrated on reaching and retaining 26–45 year-old customers. This age window represents both the highest current value and the most viable long-term customer relationship potential.

11.3 Gender: Statistically Significant but Operationally Nuanced

Male customers spend more per transaction on average (₹9,438 vs ₹8,735 for females), and this difference is highly statistically significant. However, the absolute gap is modest — approximately 8% — and should be interpreted carefully. The gender spend gap may reflect product category preference differences rather than fundamental differences in purchasing capacity or willingness to spend.

The retailer may wish to examine whether female customers are underrepresented in high-margin product categories and whether targeted product recommendations could increase female average transaction value.

11.4 City Category: Non-Metro Premium Customers

Counterintuitively, City C (the smallest city category) exhibits the highest average purchase amount (₹9,720), while City A (metro) has the lowest (₹8,912). This suggests that Black Friday spending behavior is not simply a function of urban wealth concentration.

A possible explanation is that non-metro customers, with fewer year-round access to premium products, concentrate higher-value purchases in Black Friday events. This 'event-driven spending' pattern in non-metro segments represents a strategic opportunity for targeted reach through digital and e-commerce channels.

11.5 Product Strategy Implications

The extreme concentration of revenue in Categories 1, 5, and 8 creates both a risk (over-dependence on a few categories) and an opportunity (deep investment in proven demand areas). The high HV penetration in Categories 18, 9, and 11 suggests premium product lines that disproportionately attract the most valuable customers.

A rational product strategy would balance: (1) depth investment in Category 1 to protect the revenue core, (2) premium development in Categories 18, 9, and 11 to attract and retain HV customers, and (3) rationalization of low-revenue, low-premium categories to optimize operational costs.

Chapter 12: Recommendations

12.1 Revenue Growth Strategy

Based on the finding that top product categories and premium products drive the majority of revenue:

- Expand and deepen premium product lines in Categories 18, 9, and 11, where High-Value revenue penetration exceeds 59%
- Prioritize inventory allocation and marketing spend for the top 10 revenue-generating SKUs (led by P00025442 at ₹28.0M)
- Develop an annual premium product launch calendar timed to Black Friday events to maximize top-SKU concentration
- Leverage Category 1's ₹1.91 Bn contribution as a strategic anchor — invest in its expansion as a platform category

12.2 Customer Retention & Loyalty

Given the 55.4% revenue dependence on just 20% of customers:

- Design a tiered High-Value Customer loyalty program with exclusive early access, personalized offers, and premium membership benefits
- Focus retention programs on the 26–45 age window, which contains the largest HV cohort (758 of 1,179 HV customers)
- Implement personalized re-engagement campaigns for HV customers showing declining purchase frequency
- Consider a VIP membership tier with annual revenue threshold triggers, targeting the top 1,179 customers identified in this analysis

12.3 Demographic Targeting

- Direct premium marketing and product recommendations toward the 26–35 age group as the highest-ROI demographic segment
- Develop gender-aware product recommendation models to assess whether female customers are systematically underrepresented in high-margin categories
- Invest in digital and e-commerce reach in City C and B markets, where non-metro premium spending patterns are evident
- Deprioritize marital-status-based segmentation — this variable shows no statistically significant purchasing behavior difference

12.4 Product Rationalization

With revenue highly concentrated in a minority of categories and SKUs:

- Conduct an SKU-level profitability review across low-revenue categories — consider phasing out low-volume, low-premium SKUs
- Reduce operational complexity by rationalizing Category 2 and 3 product ranges (together contributing only 9.3% of revenue)
- Reinvest operational savings from category rationalization into deepening Category 1, 5, and 8 product assortments

12.5 Dashboard & Analytics Utilization

- Operationalize the Power BI dashboard for regular executive reporting — update quarterly with fresh transaction data
- Extend the dashboard to include real-time inventory KPIs, enabling proactive product management during high-volume events
- Use the Customer Segmentation page to monitor HV customer count month-over-month as a leading revenue risk indicator

Priority Action Matrix

Immediate priority (highest ROI): (1) Establish HV loyalty program for top 1,179 customers. Short-term (3–6 months): (2) Launch premium product campaign for Categories 18, 9, 11. Medium-term (6–12 months): (3) Execute SKU rationalization for low-revenue categories. Strategic (12+ months): (4) Build predictive churn model for HV customer retention.

Chapter 13: Conclusion

13.1 Summary of the Study

This project delivered a comprehensive analytical study of Black Friday retail transaction data, successfully achieving all stated objectives. The analysis combined rigorous Python-based exploratory data analysis and statistical hypothesis testing with interactive Power BI dashboard development to produce an end-to-end business intelligence solution.

The dataset — comprising 550,068 transactions across 5,891 customers — was systematically preprocessed, statistically interrogated, and visually presented across four Power BI dashboard pages. The analysis produced quantifiable, actionable findings across customer segmentation, demographic behavior, and product performance dimensions.

13.2 Key Findings Summary

Finding	Key Metric
Revenue concentration: 20% of customers drive 55.4% of revenue	Classic Pareto effect confirmed
26–35 age group is the dominant revenue generator	₹2.03 Bn (39.9% of total revenue)
Male customers spend significantly more per transaction	₹9,438 vs ₹8,735 (Mann-Whitney $p < 0.0001$)
City C has the highest average purchase amount	₹9,720 per transaction
Marital status does not significantly affect spending	$p = 0.408$ — not significant
Category 1 alone drives 37.5% of all revenue	₹1.91 Bn out of ₹5.10 Bn total
Categories 18, 9, 11 show highest HV penetration	>59% of revenue from HV customers
Top product P00025442 generates ₹28.0 M	Flagship SKU priority established

13.3 Contribution of the Study

This project demonstrates how raw transactional data can be transformed into a coherent intelligence framework through structured analytical methodology. The combination of Python for statistical rigor and Power BI for visual communication creates a replicable template applicable to retail analytics contexts beyond Black Friday.

The segmentation framework, statistical testing protocol, and dashboard architecture developed in this study can serve as a foundation for more advanced analytical extensions including predictive modeling, customer lifetime value calculation, and real-time event analytics.

13.4 Limitations Acknowledged

- The dataset lacks time-stamp information, preventing trend analysis or event-day versus pre/post comparisons
- Occupation codes are non-descriptive, limiting the interpretability of occupation-based findings
- External factors (promotions, discounts, pricing) are absent from the dataset
- Product category labels are numeric codes without descriptive names, preventing category-level narrative enrichment

13.5 Future Scope

- Machine learning models (Random Forest, XGBoost) to predict High-Value customer classification
- Customer Lifetime Value (CLV) modeling incorporating multi-period transaction data
- Collaborative filtering-based product recommendation systems
- Real-time dashboard integration for live Black Friday event monitoring
- Cluster analysis (K-Means, DBSCAN) for unsupervised customer behavioral segmentation

13.6 Final Remarks

This analysis successfully demonstrates the power of integrated data analytics in retail intelligence. The finding that 20% of customers drive over 55% of revenue — validated through rigorous statistical testing and visualized through an interactive Power BI dashboard — provides concrete, quantified guidance for strategic investment.

The methodology, tools, and frameworks developed in this project represent a scalable approach to retail analytics that organizations can leverage to continuously sharpen their customer strategies, product decisions, and revenue optimization efforts in an increasingly data-competitive retail environment.

References & Appendices

Data Source

- Black Friday Sales Dataset — Provided by Institute for Academic Data Analytics Training
- Dataset characteristics: 550,068 transactions, 5,891 unique customers, 3,631 unique products, 12 original variables

Tools & Technologies

- Python 3.x — Primary analytical programming language
- Pandas — Data manipulation and aggregation library
- NumPy — Numerical computation library
- Matplotlib & Seaborn — Statistical visualization libraries
- SciPy — Statistical testing library (Mann-Whitney U, Kruskal-Wallis)
- Microsoft Power BI Desktop — Interactive dashboard development
- DAX (Data Analysis Expressions) — Power BI custom measure language

Statistical Methods

- Mann-Whitney U Test — Non-parametric test for two independent samples
- Kruskal-Wallis Test — Non-parametric ANOVA for multiple groups
- IQR-based Outlier Detection — Interquartile range method for outlier identification
- Percentile-based Segmentation — 80th percentile threshold for HV/LV classification
- Descriptive Statistics — Mean, median, standard deviation, quartile analysis

Documentation Resources

- Microsoft Power BI Documentation — docs.microsoft.com/power-bi
- Python Official Documentation — python.org
- Pandas Documentation — pandas.pydata.org
- SciPy Statistical Functions — docs.scipy.org

Appendix A: Dataset Column Summary

Column	Data Type	Unique Values	Missing %
User_ID	Integer	5,891	0%
Product_ID	String	3,631	0%
Gender	Categorical	2 (M, F)	0%
Age	Categorical	7 groups	0%
Occupation	Integer	21 codes	0%
City_Category	Categorical	3 (A, B, C)	0%
Stay_In_Current_City_Years	Categorical	5 values	0%
Marital_Status	Binary	2 (0, 1)	0%

Product_Category_1	Integer	20 categories	0%
Product_Category_2	Float	17 categories	31.6%
Product_Category_3	Float	15 categories	69.7%
Purchase	Integer	17,959 unique	0%